



Statistics of Financial Markets

Chapter 5: Variance Ratio Tests



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Outline

Motivation

Lo-MacKinlay Test

Chow-Denning Test

Wild Bootstrap VR Test

Applications to Stock Indices

Conclusions

References

Motivation

- ▣ Variance ratio as a tool for testing the random walk hypothesis
- ▣ Exploiting the linear variance property of random walks
- ▣ Advantages over traditional autocorrelation-based tests

Lo-MacKinlay Test

- ▣ Construction of the variance ratio statistic $VR(q)$
- ▣ Asymptotic distribution under homoskedasticity and heteroskedasticity
- ▣ Test statistics $Z(q)$ and $Z^*(q)$
- ▣ Interpretation of results

Chow-Denning Test

- ▣ Multiple variance ratio testing framework
- ▣ Controlling for size distortion across multiple holding periods
- ▣ Joint null hypothesis of random walk for all periods q
- ▣ Studentized maximum modulus distribution

Wild Bootstrap VR Test

- ▣ Motivation: improving finite-sample properties
- ▣ Wild bootstrap procedure for variance ratio tests
- ▣ Handling conditional heteroskedasticity
- ▣ Kim's (2006) wild bootstrap implementation

Applications to Stock Indices

- ▣ Empirical application to major stock market indices
- ▣ Variance ratio profiles across different holding periods
- ▣ Comparison of results across developed and emerging markets
- ▣ Python implementation and visualization

 SFM_ch5_variance_ratio

Conclusions

- ▣ Summary of variance ratio testing methodology
- ▣ Key empirical findings on market efficiency
- ▣ Strengths and limitations of variance ratio tests

References

- ▣ Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks.
- ▣ Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test.
- ▣ Kim, J.H. (2006). Wild bootstrapping variance ratio tests.